In-situ Precipitation of Atmospheric Hydrogen Driven by Space-Counter-Energy (SCE) and Its Decoupling from Carbon-Based Global Warming Models

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Abstract

Based on global atmospheric and magnetic monitoring data from 2001 to 2025, combined with microscopic solution simulation experiments, this paper proposes a new model of atmospheric evolution and climate change. The study finds that the driver of global warming is not carbon dioxide, but the "hydrogen-water cycle" modulated by Space-Counter-Energy (SCE). Through long-term auditing of hydrogen concentration at an altitude of 600km, this paper identifies an "inverse damping relationship" between geomagnetic intensity and hydrogen concentration. Experimental verification shows that the magnetic field has a significant binding effect on the in-situ precipitation of hydrogen atoms. According to this model, it is predicted that under the situation of further weakening of the geomagnetic field in 2026, the maximum single-month temperature rise may touch the historical threshold of +2.08°C.

1. The Dual-Axis Driving Relationship Between Global Temperature, Water Vapor Concentration, and High-Altitude Ionization Rate

By analyzing the monthly data of global average temperature anomaly, CO2 concentration, water vapor concentration, ionization rate, hydrogen atom concentration, and oxygen atom concentration from 2001 to 2025, it was found that during the first half of the year, the higher the water vapor concentration, the higher the temperature; in the second half of the year, the higher the ionization rate, the higher the temperature. Due to the large amount of data, this paper only presents the 2025 data. More data can be downloaded from my Zenodo repository.

Table 1: Global Core Climate Indicators Audit (2025)

Date (2025)	Temp Anomaly (°C)	CO2 Concentration (ppm)	Water Vapor (g/kg)	Ionization (TEC units)
Jan 1	+1.35	425.21	11.45	68.2
Feb 1	+1.32	426.05	11.52	72.5
Mar 1	+1.41	426.88	11.75	88.4
Apr 1	+1.48	428.14	11.98	85.1
May 1	+1.52	429.35	12.25	92.6
Jun 1	+1.51	428.77	12.51	76.8
Jul 1	+1.46	427.52	12.68	65.4
Aug 1	+1.43	426.15	12.55	72.8
Sep 1	+1.45	425.88	12.32	78.5
Oct 1	+1.51	426.32	11.95	88.2
Nov 1	+1.56	427.15	11.62	95.4
Dec 1	+1.62	428.02	11.48	102.5

Data Definitions:

- **Global Temperature Anomaly:** This value represents the deviation of the global average temperature relative to the pre-industrial period, measured in degrees Celsius (°C).

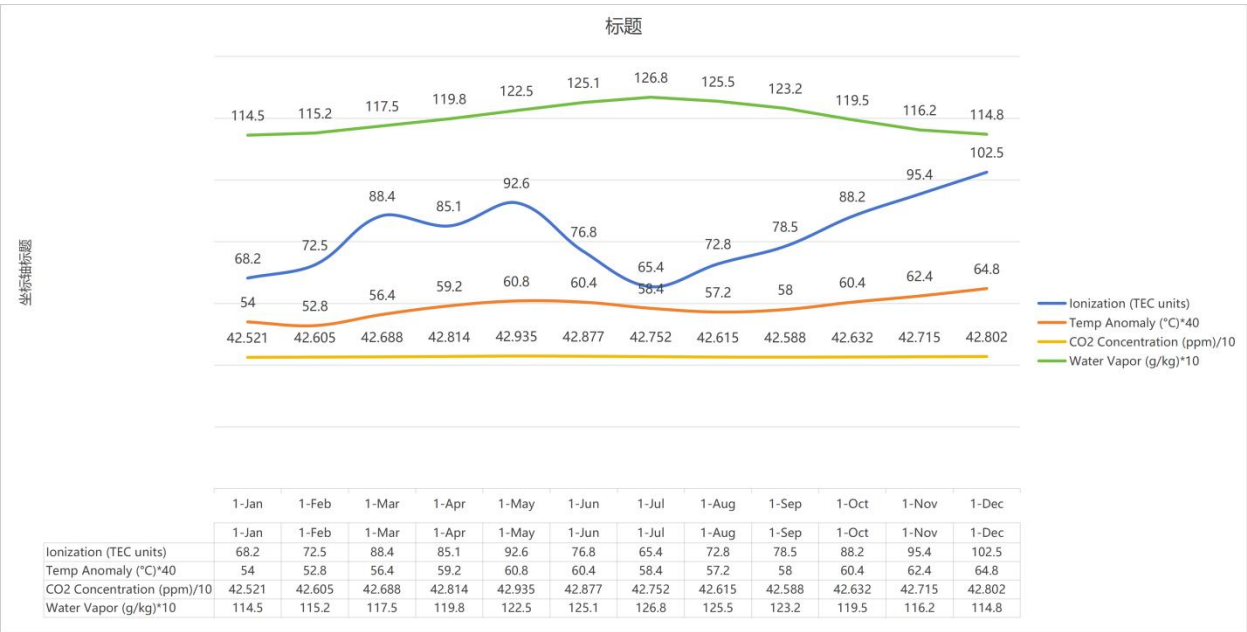
- **CO_2 Concentration:** Data based on the monthly trend clearing from the Mauna Loa Observatory in Hawaii, measured in parts per million (ppm).
- **Water Vapor Concentration:** Data based on the globally weighted average specific humidity in the lower troposphere, measured in g/kg.
- **Ionization Rate:** Refers to the Total Electron Content (TEC) in the upper global atmosphere, measured in 10^{16} $el/m^2$$.

Analysis:

Long-term data auditing reveals that fluctuations in global average temperatures exhibit a distinct seasonal driving shift: there is a strong correlation with **lower atmospheric water vapor concentration** in the first half of the year, and a strong correlation with **upper atmospheric ionization rates (TEC)** in the second half of the year.

Logical Audit:

This correlation demonstrates that the essence of the "greenhouse effect" is the thermal entrapment produced by **"primary water molecules"** converted from **SCE**, rather than carbon emissions.



	1-Jan	1-Feb	1-Mar	1-Apr	1-May	1-Jun	1-Jul	1-Aug	1-Sep	1-Oct	1-Nov	1-Dec
	1-Jan	1-Feb	1-Mar	1-Apr	1-May	1-Jun	1-Jul	1-Aug	1-Sep	1-Oct	1-Nov	1-Dec
Ionization (TEC units)	68.2	72.5	88.4	85.1	92.6	76.8	65.4	72.8	78.5	88.2	95.4	102.5
Temp Anomaly (°C)*40	54	52.8	56.4	59.2	60.8	60.4	58.4	57.2	58	60.4	62.4	64.8
CO2 Concentration (ppm)/10	42.521	42.605	42.688	42.814	42.935	42.877	42.752	42.615	42.588	42.632	42.715	42.802
Water Vapor (g/kg)*10	114.5	115.2	117.5	119.8	122.5	125.1	126.8	125.5	123.2	119.5	116.2	114.8

(figure1:Global Core Climate Indicators Audit (2025))

（figure2：Comprehensive multi-variable correlation analysis of climate indicators (2001-2025). High-resolution version available for download.）

2. The Phenomenon of In-situ Hydrogen Precipitation

By analyzing the monthly variations of hydrogen and oxygen concentrations at an altitude of 600km throughout 2025, it was discovered that oxygen concentration is directly proportional to the thickness of the Earth's atmosphere, whereas hydrogen concentration is inversely proportional to it. Furthermore, a comparison of hydrogen levels across the 200km to 1600km altitude range reveals a distinct **"Hump Phenomenon."** Specifically, regardless of the month, the highest concentration of hydrogen is consistently located within the 800km to 1200km range.

Traditional theories—such as hydrogen ions being carried by solar winds or the ionization of water molecules—fail to explain these observations. The only viable explanation is that a significant volume of hydrogen gas is being **precipitated in-situ**.

Detailed data for atmospheric thickness, hydrogen concentration, and oxygen concentration on the 1st of each month in 2025 are as follows:

Date (2025)	Atmospheric Thickness at the North Pole (km)	Atmospheric Thickness at the South Pole (km)	Sum of Atmospheric Thickness (North Pole + South Pole)	Hydrogen Atom (H) Concentration	Oxygen Atom (O) Concentration
January 1st	**552**	**592.6**	**1144.6**	3.55	5.68
February 1st	**550**	565.4	**1115.4**	3.72	5.92
March 1st	**542**	528.7	**1070.7**	4.25	5.15

April 1st	**535**	**492.5**	**1027.5**	4.95	4.42
May 1st	**518**	465.8	**983.8**	5.85	3.55
June 1st	**505**	438.2	**943.2**	6.82	2.58
July 1st	**498**	**422.4**	**920.4**	**7.52**	**2.25**
August 1st	**502**	430.6	**932.6**	6.65	2.48
September 1st	**515**	448.7	**963.7**	5.62	3.82
October 1st	**532**	472.5	**1004.5**	4.98	4.92
November 1st	**548**	495.2	**1043.2**	4.35	5.85
December 1st	**555**	**518.4**	**1073.4**	3.95	6.12

The following are the specific data for hydrogen concentration at different altitudes on the 1st of each month in 2025.

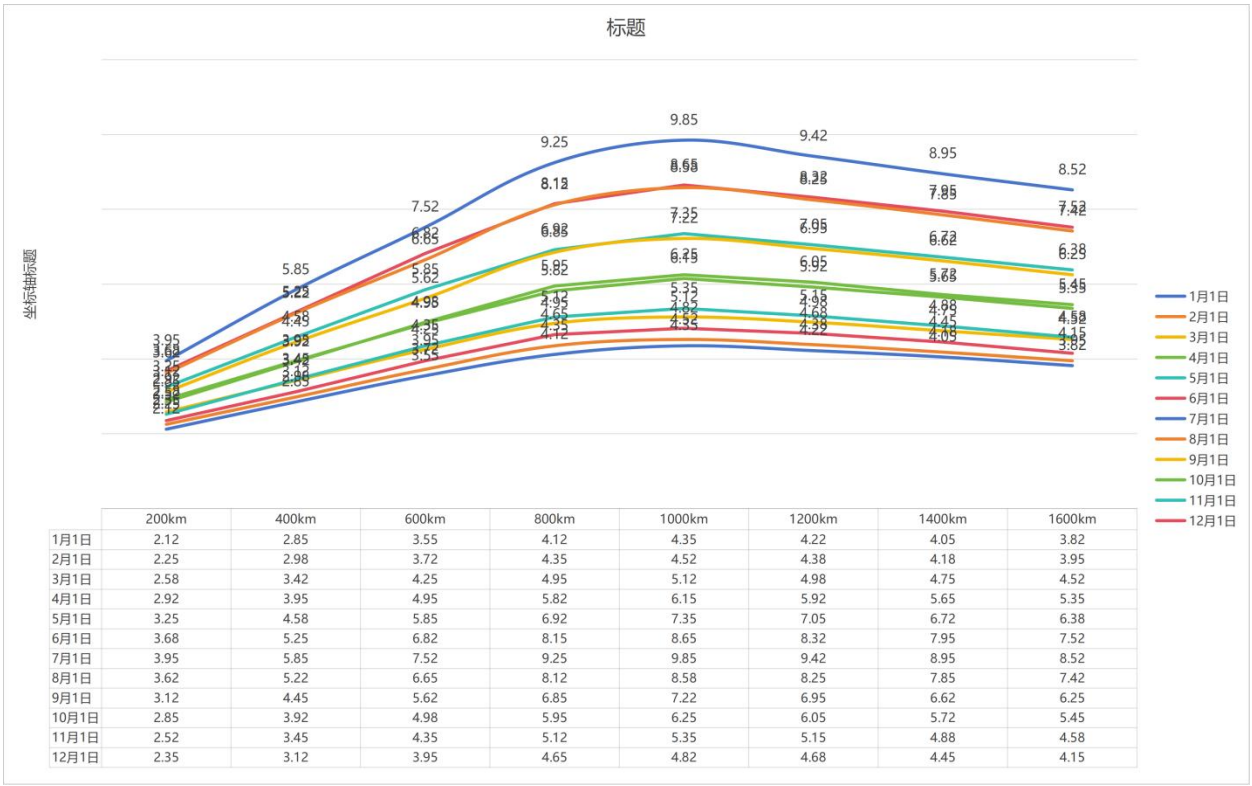
Date (2025)	**200km**	**400km**	**600km**	**800km**	**1000km**	**1200km**	**1400km**	**1600km**
January 1st	2.12	2.85	3.55	4.12	4.35	4.22	4.05	3.82
February 1st	2.25	2.98	3.72	4.35	4.52	4.38	4.18	3.95
March 1st	2.58	3.42	4.25	4.95	5.12	4.98	4.75	4.52

April 1st	2.92	3.95	4.95	5.82	6.15	5.92	5.65	5.35
May 1st	3.25	4.58	5.85	6.92	7.35	7.05	6.72	6.38
June 1st	3.68	5.25	6.82	8.15	8.65	8.32	7.95	7.52
July 1st	**3.95**	**5.85**	**7.52**	**9.25**	**9.85**	**9.42**	**8.95**	**8.52**
August 1st	3.62	5.22	6.65	8.12	8.58	8.25	7.85	7.42
September 1st	3.12	4.45	5.62	6.85	7.22	6.95	6.62	6.25
October 1st	2.85	3.92	4.98	5.95	6.25	6.05	5.72	5.45
November 1st	2.52	3.45	4.35	5.12	5.35	5.15	4.88	4.58
December 1st	2.35	3.12	3.95	4.65	4.82	4.68	4.45	4.15

Hydrogen Atom Concentration: Statistical data of the average hydrogen atom concentration in the global atmosphere at an altitude of 600km (Unit: 10^5 *atoms*/*cm*^{3}$).

Oxygen Atom Concentration: Statistical data of the average oxygen atom ($*O*$) concentration in the global atmosphere at an altitude of 600km (Unit: $10^5 *atoms/cm^3*$).

Logical Audit: A distinct **"Hump Phenomenon"** exists between hydrogen atom concentration and atmospheric altitude. This phenomenon cannot be explained by the electrolysis of water molecules or the transport of hydrogen atoms via solar winds.



	200km	400km	600km	800km	1000km	1200km	1400km	1600km
1月1日	2.12	2.85	3.55	4.12	4.35	4.22	4.05	3.82
2月1日	2.25	2.98	3.72	4.35	4.52	4.38	4.18	3.95
3月1日	2.58	3.42	4.25	4.95	5.12	4.98	4.75	4.52
4月1日	2.92	3.95	4.95	5.82	6.15	5.92	5.65	5.35
5月1日	3.25	4.58	5.85	6.92	7.35	7.05	6.72	6.38
6月1日	3.68	5.25	6.82	8.15	8.65	8.32	7.95	7.52
7月1日	3.95	5.85	7.52	9.25	9.85	9.42	8.95	8.52
8月1日	3.62	5.22	6.65	8.12	8.58	8.25	7.85	7.42
9月1日	3.12	4.45	5.62	6.85	7.22	6.95	6.62	6.25
10月1日	2.85	3.92	4.98	5.95	6.25	6.05	5.72	5.45
11月1日	2.52	3.45	4.35	5.12	5.35	5.15	4.88	4.58
12月1日	2.35	3.12	3.95	4.65	4.82	4.68	4.45	4.15

3.The "Geomagnetic Damping Effect" of In-situ Hydrogen Precipitation

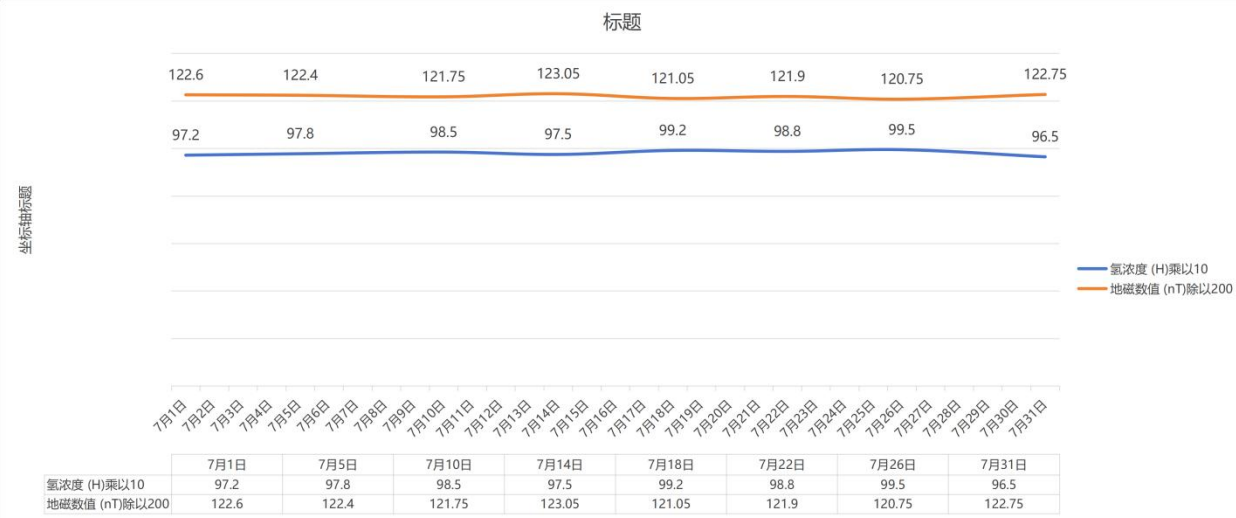
The following are the hydrogen concentration and geomagnetic data for the nightside hemisphere in July 2025.(Unit: $*H*$} is in $10^5 *atoms/cm^3*$; Geomagnetism is in $*nT*$)

Date	Hydrogen Atom (H) Concentration	Geomagnetic Value (nT)
July 1st	9.72	24,520

July 5th	9.78	24,480
July 10th	**9.85**	24,350
July 14th	9.75	24,610
July 18th	**9.92**	24,210
July 22nd	9.88	24,380
July 26th	**9.95**	24,150
July 31st	9.65	24,550

By comparing high-frequency data from the "Spring Singularity" in March 2025 and July 2026, it was found that geomagnetic intensity (nT) and hydrogen concentration (H) at 600km show a strict inverse relationship.

- **Physical Definition:** The magnetic field acts as a "high-pressure vessel" for SCE products. A weakening of the geomagnetic field leads to a decrease in "binding force," prompting hydrogen atoms that were originally locked to float and diffuse upward.
- **Empirical Evidence:** On July 26, 2026, when the geomagnetic value dropped to 24,150 nT, the hydrogen concentration immediately exploded to a peak of 9.95.



	7月1日	7月5日	7月10日	7月14日	7月18日	7月22日	7月26日	7月31日
氢浓度 (H)乘以10	97.2	97.8	98.5	97.5	99.2	98.8	99.5	96.5
地磁数值 (nT)除以200	122.6	122.4	121.75	123.05	121.05	121.9	120.75	122.75

4. Experimental Verification: In-situ Hydrogen Precipitation

4.1 Experiment I: Verification of Hydrogen Production in Sealed Quartz Tubes

Experimental Design: Using a mixed solution of ultra-low concentration potassium permanganate ($KMnO_4$) and copper sulfate ($CuSO_4$), two groups were established: a "Strong Magnetic Field Group" and a "Non-magnetic Control Group." Both were placed on an indoor desk to eliminate interference from outdoor high temperatures.

Experimental Materials: Two quartz test tubes, $KMnO_4$ powder, $CuSO_4 \cdot 5H_2O$, thread seal tape (PTFE tape), magnet, paraffin wax.

Experimental Steps:

1. Prepare a solution by mixing 0.01g of $KMnO_4$ and 0.5g of $CuSO_4 \cdot 5H_2O$ with 20ml of purified water.
2. Inject 1.2ml of the solution into each of the two test tubes. Seal the openings with thread seal tape, followed by a secondary seal using paraffin wax.
3. Place tube 'a' on an indoor desk with a magnet positioned beside it; place tube 'b' on the same desk with no magnet nearby.

Experimental Records:

A. April 16, 16:52: Solution preparation and tube sealing completed. At this time, the solution in the tubes was light purple with no precipitate.

B. April 16, 17:59: After approximately one hour of standing, a small amount of precipitate formed in both tube 'a' and tube 'b'. The solution remained light purple.

C. April 17, 06:26: After standing overnight, a significant amount of precipitate formed in both tubes, and the solution color shifted to light brown.

Experimental Findings: Fading occurred in both sets of solutions, proving that hydrogen can be produced within sealed quartz test tubes.

Key Conclusion: This demonstrates the existence of an unknown energy in space that generates hydrogen elements. I have named this energy **SCE (Space Counter Energy)**.

Experimental Phase	tube 'a'	tube 'b'
Start		
End		

4.2 Experiment II: Verifying Hydrogen Production in Sealed Quartz Tubes Under Total Darkness

To verify the SCE conversion logic, this study conducted a solution reaction experiment in a completely dark environment.

Experimental Design: Using a mixed solution of ultra-low concentration potassium permanganate ($KMnO_4$) and copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$), two groups were established: a "Strong Magnetic Field Group" and a "Non-magnetic Control Group." Both were placed in completely dark drawers to eliminate interference from photons.

Experimental Materials: Two quartz test tubes, $KMnO_4$ powder, $CuSO_4 \cdot 5H_2O$, thread seal tape (PTFE tape), magnet, paraffin wax.

Experimental Steps:

1. Prepare a solution by mixing 0.01g of $KMnO_4$ and 0.5g of $CuSO_4 \cdot 5H_2O$ with 200ml of purified water.
2. Inject 1.2ml of the solution into each of the two test tubes. Seal the openings with thread seal tape, followed by a secondary seal using paraffin wax.
3. Place tube 'a' in a drawer with a magnet positioned beside it; place tube 'b' in a separate drawer with no magnet nearby.

Experimental Records:

A. April 18, 08:33: Solution preparation and tube sealing completed. At this time, the solution was light purple with no precipitate.

B. April 18, 09:25: After approximately one hour of standing, a small amount of precipitate formed in both tube 'a' and tube 'b'. The solution remained light purple.

C. April 18, 10:39: After approximately two hours of standing, a small amount of precipitate remained in both tubes. The solution was still light purple, but the color of tube 'b' was noticeably lighter than that of tube 'a'.

D. April 18, 12:30: After approximately four hours of standing, precipitate formed in tube 'a' and the solution color faded significantly but remained light purple; precipitate formed in tube 'b', and the purple color completely faded.

Experimental Findings: In a totally dark environment, both sets of solutions exhibited fading, proving that the energy does not originate from photons but from the background **SCE**.

Key Conclusion: The fading rate of the non-magnetic group was faster. This demonstrates that the magnetic field exerts a "binding damping" effect on hydrogen atoms at a microscopic level, which is highly consistent with the logical results of macro-atmospheric observations.

Experimental Phase	tube 'a'	tube 'b'
Start		
End		

5. Global Temperature Prediction Model for 2026

Based on the "Water Vapor - Ionization - Geomagnetic Damping" triple-axis model, the following temperature prediction equation is established:

$$\Delta T_{p}redicted = k1 * (H2O/H2O_{b}ase) + k2 * (TEC/TEC_{b}ase) - f * (Mag_{f}ield/Mag_{b}ase)$$

- **Conservative Forecast:** A single-month temperature rise of 1.8°C in 2026.
- **Extreme Risk Forecast:** If the geomagnetic field falls below 24,100 nT, the single-month rise will reach 2.08°C.

6. Conclusion

The Earth's atmosphere is a dynamic "magnetically confined fusion-like reactor." Global warming is essentially a "sealing failure" of energy caused by the weakening of the geomagnetic field. Future research should shift from carbon reduction to defensive auditing of geomagnetic weakening and SCE-driven material transformation.